

Dr. Daniel López-Cano

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- ACADEMIC APPOINTMENTS

11/24 –
Present



FAPESP Postdoctoral Research Fellow – *Institute of Physics, University of São Paulo (IFUSP)* · São Paulo, Brazil 

Supervisor: Prof. Luis Raul Weber Abramo.

Topics: cosmological simulations, semi-supervised machine learning techniques , research/teaching: Lyman- α processes, differential geometry.

02/26 – 06/26



FAPESP-BEPE Visiting Postdoctoral Research Fellow – *Institut de Física d'Altes Energies (IFAE)* · Cerdanyola del Vallès / Barcelona, Spain 

Host supervisor: Prof. Andreu Font-Ribera. Co-mentor: Dr. Jonás Chaves-Montero.

Project: *SkewerJAX: fast Ly- α skewers and a survey-ready metal-contamination model.*

- EDUCATION

09/20 – 09/24



PhD in Astrophysics – *Donostia International Physics Center (DIPC) / Universidad Autónoma de Madrid (UAM)* · Donostia-San Sebastián / Madrid, Spain 

Supervisor: Prof. Raúl E. Angulo. PhD programme in Astrophysics at UAM.

Thesis: “Improving cosmological simulations: semi-analytic models, the internal structure of haloes, and machine learning techniques”. 

Topics: N-body simulations, structure formation & machine learning , simulation based inference , emulators (BACCO ).

09/19 – 07/20



MSc in Theoretical Physics – *Institute of Theoretical Physics IFT UAM-CSIC* · Madrid, Spain 

Specialisation in Astrophysics and Physics of the Cosmos. Supervisor: Prof. Alexander Knebe.

Thesis: “UNITSIM-Galaxies: data release and clustering of emission-line galaxies”.

09/15 – 07/19



BSc in Physics – *Universidad Autónoma de Madrid* · Madrid, Spain 

Theoretical thesis: “UNIT-SAGE galaxies: studying galaxy statistics using cosmological simulations”. Supervisor: Prof. Alexander Knebe.

Experimental thesis: “Tight-binding method and evolutionary algorithms”. Supervisor: Prof. Juan José Palacios Burgos.

- ADDITIONAL R&D / INDUSTRY EXPERIENCE

04/19 – 07/19



R&D Software Developer Intern, GNSS – *GMV* · Tres Cantos, Madrid, Spain
Global Navigation Satellite Systems department. 

Topics: Python development for precise point positioning data analysis and scientific software workflows.

- RESEARCH GRANTS AND FELLOWSHIPS

 **FAPESP Postdoctoral Fellowship**

São Paulo Research Foundation (FAPESP), Brazil. Process: **2024/05768-9**.

Project: *Connecting galaxies and dark-matter haloes*. Hosted at IFUSP.

11/2024 – 10/2026

 **FAPESP-BEPE Fellowship for Research Internship Abroad**

São Paulo Research Foundation (FAPESP), Brazil. Process: **2025/24623-4**.

Project: *SkewerJAX: fast Ly- α skewers and a survey-ready metal-contamination model*. Hosted at IFAE, Barcelona.

02/2026 – 06/2026

Teaching & Supervision

- 2025** *From Differential Geometry to General Relativity*
Instructor 08/09/2025–06/10/2025 · IFUSP, São Paulo, Brazil
Graduate course, 3 credits; 18h lectures + 9h practical sessions.
- 2026** *Machine Learning and Neural Networks for Physics*
Instructor 20/07/2026–24/07/2026 · 1st IFUSP Winter School, São Paulo, Brazil
5 lectures / 1.5h each. ML foundations + hands-on exercises.
- 2024–2026** MSc Co-advisor: Victor Roberto Soares da Silva
Supervision 10/2024–07/2026 · IFUSP, São Paulo, Brazil
Co-supervised with Prof. Luis Raul Weber Abramo.
Project area: cosmological simulations and Ly- α systems.

Educational & Pedagogical Resources

- 2026** *From Differential Geometry to General Relativity*
Notes/code In preparation · Open teaching material
Graduate-level lecture notes combining formal derivations, examples, and visual intuition; accompanied by Python resources for figures, animations, interactive examples, and exercises.  .
- 2026** *Notes on Ly- α forest and Absorption Processes in Cosmology*
Notes/code In preparation · Pedagogical research material
Structured notes on Ly- α skewer generation, optical depth, transmitted flux, velocity-space mappings, and metal-line modelling; accompanied by JAX code for reproducible examples and exercises.  .

Invited Talks & Seminars

- 2026** DFT & CIAFF / DeCAF Seminar
Seminar 25/02/2026 · Universidad Autónoma de Madrid, Spain
- 2025** WEAVE Collaboration Meeting
Invited 01/09/2025–05/09/2025 · University of Milano-Bicocca, Milan, Italy
- 2025** IFAE Seminar 
Seminar 14/04/2025 · Institut de Física d'Altes Energies, Barcelona, Spain
- 2024** São Paulo Group Meetings in Astro & Cosmo 
Invited 08/11/2024 · IFT-UNESP / Instituto Principia, São Paulo, Brazil
- 2024** Agorá Meeting DFMA 
Invited 25/10/2024 · Instituto de Física, University of São Paulo, Brazil

Research Visits & Scientific Stays

- 2023** Simons Foundation 
Visit 31/07/2023–20/08/2023 · Visiting Research Stay during PhD. Center for Computational Mathematics, Flatiron Institute, NY, USA

- 2022** Munich Institute for Astro-, Particle and BioPhysics 
Program 09/05/2022–03/06/2022 · Advances in Cosmology through Numerical Simulations. Garching's Research Center, Germany

Conferences, Workshops & Schools

- 2025** 8th ICE Summer School 
School 07/06/2025–18/06/2025 · ICE-CSIC, Barcelona, Spain
- 2024** Fundamental Physics Meets Current and Future Facilities in Cosmology 
Workshop 02/12/2024–13/12/2024 · ICTP-SAIFR, São Paulo, Brazil
- 2024** Cosmology with the Large-Scale Structure of the Universe
Poster 22/05/2024–24/05/2024 · DIPC, Donostia / San Sebastián, Spain
- 2023** Cosmic Connections: A Machine Learning × Astrophysics Symposium
Contributed 15/05/2023–18/05/2023 · Center for Computational Astrophysics, Flatiron Institute, NY, USA
- 2022** VII Meeting on Fundamental Cosmology
Contributed 02/11/2022–04/11/2022 · Institute of Astrophysics of Andalusia, Spain
- 2021** 18th MultiDark Consolider Workshop
Contributed 18/10/2021–20/10/2021 · International University of Andalusia, Spain

Computational & Technical Skills

- Programming**
Python, C, C++, Fortran, Unix shell / Bash, MATLAB.
- Scientific Python & machine learning**
JAX, PyTorch, TensorFlow/Keras, NumPy, SciPy, Pandas, Matplotlib, scikit-learn, Weights & Biases.
- Cosmological simulations & survey modelling**
N-body simulations, Gadget/L-Gadget, mock-catalogue construction, synthetic observables, Ly- α skewer generation, photometric and spectroscopic survey data.
- HPC & reproducibility**
Linux, SLURM, HTCondor, MPI workflows, OpenMP parallelisation, Git/GitHub, Jupyter, LaTeX.

Professional Training

- 2023** TensorFlow Specialization: Custom Models, Layers, and Loss Functions
Course 27/12/2022–24/01/2023 · Coursera, online
- 2021** Machine Learning
Course 05/07/2021–09/09/2021 · Coursera, online
- 2021** Scientific Writing and Presentation
Training 09/06/2021–10/06/2021 · Institute of Astrophysics of Andalusia, online
- 2020** XV Spanish Virtual Observatory School
School 06/02/2020 · Autonomous University of Madrid, Spain

Outreach & Public Engagement

- 2023** Astronomical Exhibition Guide: STROM – Inclusive Astronomy 
Guide 04/10/2023–31/10/2023 · Tabakalera, Donostia / San Sebastián, Spain

Additional Scientific Experience

- 2020** **Optical Observations of GRB 200215A with CAFOS** 
Observing 17/02/2020 · Calar Alto Observatory, Spain
- 2020** **Spectroscopy and Photometry of Arp 269 with CAHA-CAFOS**
Observing 16/02/2020 · Calar Alto Observatory, Spain

Languages

Languages
Spanish: native; English: C2; Portuguese: basic working proficiency.

- PUBLICATIONS

- 2026** **J-PAS: Semi-Supervised Sim-to-Obs Transfer for Robust Star–Galaxy–Quasar Classification** **First author**
Daniel López-Cano, L. Raul Abramo, L. Nakazono, I. Pérez-Ràfols, et al.
arXiv preprint, arXiv:2602.13902, doi:10.48550/arXiv.2602.13902 
- 2026** **OJALÁ: Optimizing J-PAS Astronomy for Large-scale Analysis. A foundation model for the SED of galaxies, QSOs and stars** **Collaboration**
G. Martínez-Solaache, R. M. González Delgado, R. García-Benito, . . . , D. López-Cano, et al.
arXiv preprint, arXiv:2604.00661, doi:10.48550/arXiv.2604.00661 
- 2024** **Characterizing structure formation through instance segmentation** **First author**
Daniel López-Cano, Jens Stücker, Marcos Pellejero Ibañez, Raúl E. Angulo, Daniel Franco-Barranco
Astronomy & Astrophysics, 685, A37 
- 2024** **An improved halo occupation distribution prescription from UNITsim H_α emission-line galaxies: conformity and modified radial profile** **Co-author**
Guillermo Reyes-Peraza, Santiago Avila, Violeta Gonzalez-Perez, Daniel López-Cano, Alexander Knebe, Sujatha Ramakrishnan, Gustavo Yepes
Monthly Notices of the Royal Astronomical Society, 529, 3877–3893 
- 2022** **The cosmology dependence of the concentration–mass–redshift relation** **First author**
Daniel López-Cano, Raúl E. Angulo, Aaron D. Ludlow, M. Zennaro, S. Contreras, Jonás Chaves-Montero, G. Aricò
Monthly Notices of the Royal Astronomical Society, 517, 2000–2011 
- 2022** **UNITSIM-Galaxies: data release and clustering of emission-line galaxies** **Second author**
Alexander Knebe, Daniel López-Cano, Santiago Avila, Ginevra Favole, Adam R. H. Stevens, Violeta Gonzalez-Perez, Guillermo Reyes-Peraza, Gustavo Yepes, Chia-Hsun Chuang, Francisco-Shu Kitaura
Monthly Notices of the Royal Astronomical Society, 510, 5392–5407 